

## 5V, 2A Current Mode Synchronous Step-Down DC/DC Converter

### Description

ME3120 is a current mode synchronous step-down DC/DC converter with an input voltage range of 2.5V to 6V. And its output voltage is from 0.6V to the input voltage.

It works at PFM/PWM mode, it operates in nearly constant switching frequency 1.5MHz (TYP) over the heavy load and works in PFM mode that maintains high efficiency during light load operation.

ME3120 integrated a few protect circuits to make the system works in safety stations. It contains input Under-Voltage Lockout,  $V_{IN}$  Over-Voltage protection, over-current protection and short-circuit protection and over temperature protection. In addition, there is also a fast discharge output function

### Feature

- Output current: 2A(TYP)
- Efficiency: 95% (@ $I_{OUT}=500mA$ ,  $V_{OUT}=3.3V$ )
- Input Voltage: 2.5V~6V
- Feedback Voltage: 0.6V ( $\pm 2.0\%$ ) Allows Low  $V_{OUT}$
- Frequency: 1.5MHz (TYP) at PWM model
- PFM/PWM control mode
- Low Dropout Operation: 100% Duty Cycle
- Input Voltage UVLO& OVP
- Internally Soft-Start Function
- Output fast discharge function
- Over-current protection and short-circuit protection
- Over Temperature protection

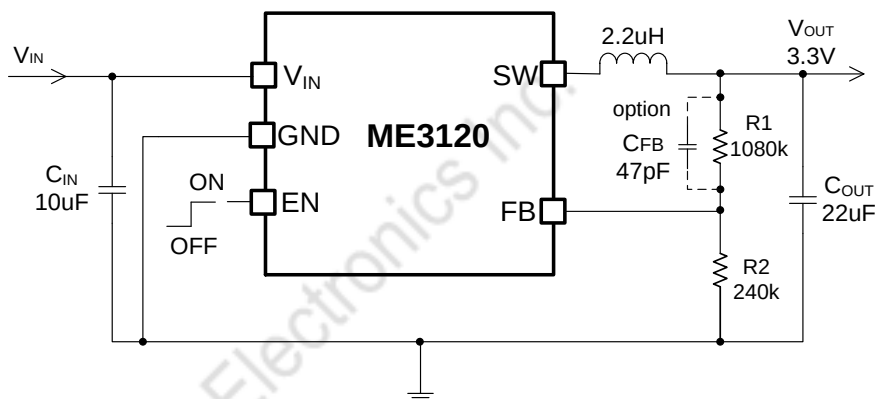
### Applications

- Bluetooth, Cable Modem
- LCD TV Power Supply& Metering Platforms
- General Purpose Point-of-Load(POL)
- Mobile devices

### Package

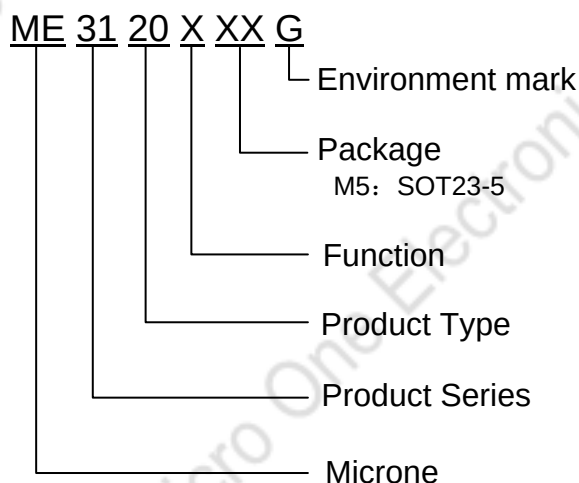
- 5-pin SOT23-5

## Typical Application Circuit



Note: Using  $C_{FB}$  can achieve better load transient response effect.

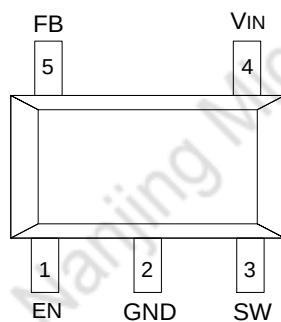
## Selection Guide



| product serie | product description |
|---------------|---------------------|
| ME3120AM5G    | Package:SOT23-5     |

**NOTE:** If you need other voltage and package, please contact our sales staff.

## Pin Configuration

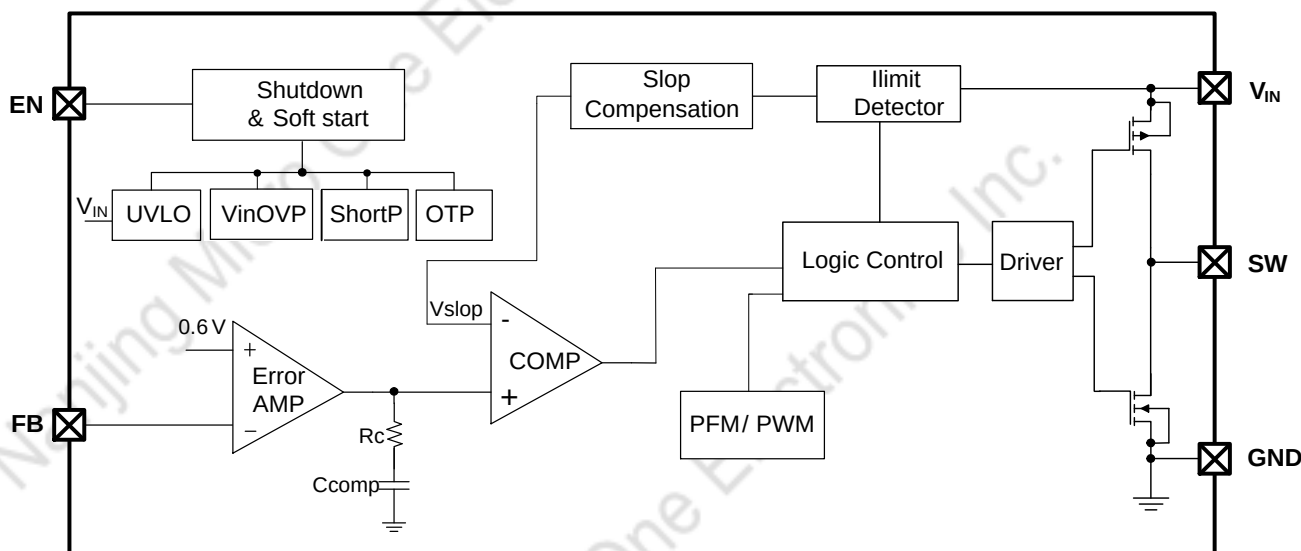


**SOT23-5**

## Pin Assignment

| PIN Number | symbol          | Function |
|------------|-----------------|----------|
| 1          | EN              | Enable   |
| 2          | GND             | Ground   |
| 3          | SW              | Switch   |
| 4          | V <sub>IN</sub> | Input    |
| 5          | FB              | Feedback |

## Block Diagram



## Absolute Maximum Ratings

| Parameter                                      | Symbol           | Ratings  | Units |
|------------------------------------------------|------------------|----------|-------|
| Input Voltage                                  | V <sub>IN</sub>  | -0.3~6.0 | V     |
| SW Pin Voltage                                 | V <sub>SW</sub>  | -0.3~6.0 | V     |
| EN Pin Voltage                                 | V <sub>EN</sub>  | -0.3~6.0 | V     |
| FB Pin Voltage                                 | V <sub>FB</sub>  | -0.3~6.0 | V     |
| Internal Power Dissipation (SOT23-5)           | P <sub>d</sub>   | 600      | mW    |
| Thermal resistance (Junction to air) (SOT23-5) | θ <sub>JA</sub>  | 210      | °C/W  |
| Operating Temperature Range                    | T <sub>Opr</sub> | -40~+105 | °C    |
| Storage Temperature Range                      | T <sub>stg</sub> | -55~+150 | °C    |
| Maximum junction temperature                   | T <sub>J</sub>   | -40~+150 | °C    |

### Notes:

- 1) Stresses at or above those listed under Absolute Maximum Ratings may cause permanent damage to the product.
- 2) The maximum allowable power dissipation is a function of the maximum junction temperature T<sub>J(MAX)</sub>, the junction-to- ambient thermal resistance θ<sub>JA</sub>, and the ambient temperature T<sub>A</sub>. The maximum allowable continuous power dissipation at any ambient temperature is calculated by P<sub>D(MAX)</sub>=(T<sub>J(MAX)</sub>-T<sub>A</sub>)/θ<sub>JA</sub>.
- 3) The θ<sub>JA</sub> values given in this table are for comparison with other packages only and cannot be used for design purposes. They do not represent the performance achieved in real-world applications.

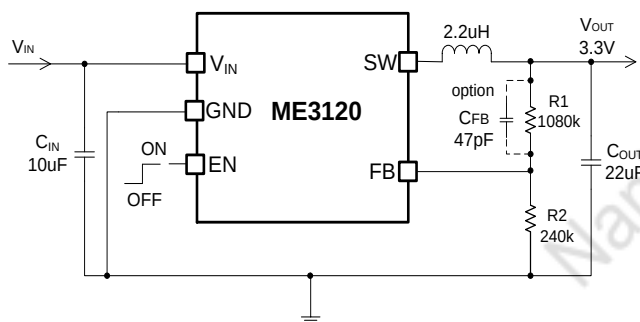
## Electrical Characteristic

ME3120 test conditions:  $V_{IN}=5V$ ,  $V_{EN}=5V$ ,  $T_{opt}=25^{\circ}C$  unless otherwise noted.

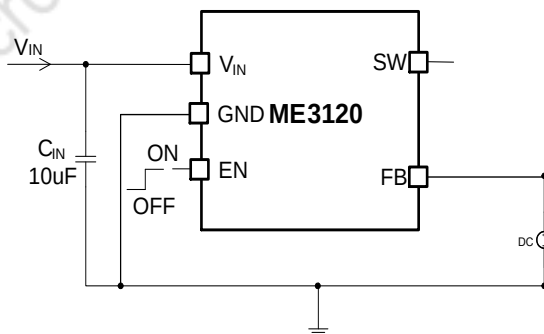
| Parameter                   | Symbol                | Condition                                                           | Min   | Typ   | Max   | Unit        | CIRCUIT |
|-----------------------------|-----------------------|---------------------------------------------------------------------|-------|-------|-------|-------------|---------|
| Input Voltage Range         | $V_{IN}$              |                                                                     | 2.5   | --    | 6     | V           | 1       |
| Feedback Voltage            | $V_{FB}$              |                                                                     | 0.588 | 0.6   | 0.612 | V           | 1       |
| VIN Over-Voltage protection | $V_{INOV}$            | $V_{IN}$ rising                                                     | 6.0   | 6.15  | 6.5   | V           | 1       |
|                             |                       | Hysteresis                                                          | --    | 0.1   | --    | V           | 1       |
| UVLO Voltage                | $V_{UVLO}$            | $V_{IN}$ rising                                                     | 2     | 2.3   | 2.45  | V           | 1       |
|                             |                       | Hysteresis                                                          | --    | 0.14  | --    | V           | 1       |
| VOUT Line Regulation        | LNR                   | $V_{IN}=2.5V$ to $5.5V$<br>$I_{OUT}=10mA$                           | --    | --    | 0.6   | %/V         | 1       |
| VOUT Load Regulation        | LDR                   | $0mA < I_{OUT} < 800mA$                                             | --    | --    | 1     | %/V         | 1       |
| DC Bias Current             | $I_q$                 | $V_{FB}=1V$ , $V_{IN}=V_{EN}=5V$                                    | --    | 50    | 65    | $\mu A$     | 2       |
|                             | $I_{SBY}$             | $V_{EN}=0V$ , $V_{IN}=5V$                                           | --    | 0     | 1     | $\mu A$     | 2       |
| Switching Frequency         | $F_{OSC}$             | $V_{IN}=5V$ , $I_{OUT}=300mA$                                       | 1.2   | 1.5   | 1.8   | Mhz         | 1       |
| Over-Temperature Threshold  | $T_{SD}$              | $I_{OUT}=10mA$                                                      | --    | 160   | --    | $^{\circ}C$ | 1       |
|                             | $T_{SDHY}$            | $I_{OUT}=10mA$                                                      | --    | 30    | --    | $^{\circ}C$ | 1       |
| PMOS ON Resistance          | $R_{PMOS}$            | $V_{FB}=0V$ , $I_{LX}=100mA$                                        | --    | 0.110 | --    | $\Omega$    | 2       |
| NMOS ON Resistance          | $R_{NMOS}$            |                                                                     | --    | 0.085 | --    | $\Omega$    |         |
| Peak Current Limit          | $I_{LIMH}$            |                                                                     | --    | 2.8   | --    | A           | 1       |
| Enable Input Voltage        | $V_{ENH}$             | $V_{FB}=0V$ , Applied voltage to $V_{EN}$ , $V_{SW}$ up to $V_{IN}$ | 1.3   | --    | 6     | V           | 2       |
|                             | $V_{ENL}$             | $V_{FB}=0V$ , Applied voltage to $V_{EN}$ , $V_{SW}$ down to GND    | GND   | --    | 0.5   | V           | 2       |
| Soft-start Time             | tss                   | $V_{IN}$ is added to the EN until the $V_{OUT}$ setup               | 0.1   | 0.3   | 0.5   | mS          | 1       |
| DISCHARG                    | $R_{DISCHA}$<br>$R_G$ | $V_{EN}=0V$ , $V_{IN}=5V$ , $V_{SW}=4V$                             | --    | 220   | --    | $\Omega$    | 2       |

## Test Circuits

Circuit No.1



Circuit No.2



## Operation Principles

The ME3120 is a synchronous low voltage step-down DC/DC converter that can support the input voltage range of 2.5V to 6V and the output voltage could low to 0.6V.

The error amplifier EA adjusts COMP voltage by comparing the feedback signal ( $V_{FB}$ ) from the output voltage with the internal reference. When the load current increases, it causes a drop in the feedback voltage relative to the reference, then the COMP voltage rises to allow higher inductor current to match the load current.

### Soft start

An internal current source charges an internal capacitor to build the soft-start ramp voltage. The  $V_{FB}$  voltage will track the reference voltage during soft-start interval. The typical soft-start time is 0.3ms.

### PFM/PWM Auto change

In PFM/PWM control mode, the ME3120 works in two states according to the load condition. It adopts PFM mode in light load, PWM mode in heavy load. So the ME3100 could acquire high efficiency in all over the loads.

### Enable

A logic-high enables the converter; a logic-low forces the device into shutdown mode.

### UVLO

If the  $V_{IN}$  lower than threshold voltage 2.15V (TYP), the UVLO comparator's output will go high and the switch controller will turn off the high side MOSFET. If the  $V_{IN}$  higher than 2.3V (typical value), the controller will resume the working state.

### $V_{IN}OVP$

If the  $V_{IN}$  higher than threshold voltage 6.15V (TYP), the chip triggers overvoltage protection function and the switch controller will turn off the high side MOSFET. If the  $V_{IN}$  lower than 6.0V (typical value), the controller will resume the working state.

### Over temperature protection (OTP)

The ME3120 implements an internal thermal shutdown function when the junction temperature exceeds 160°C (TYP). The thermal shutdown forces the device to stop switching when the junction temperature exceeds the thermal shutdown threshold. Once the temperature decreases below the hysteresis of 30°C (TYP), the device works again.

### 100% Duty Cycle Operation

When the input supply voltage decreases toward the output voltage, the duty cycle increases toward the maximum on-time. Further reduction of the supply voltage forces the main switch to remain on for more than one cycle, eventually reaching 100% duty cycle. The output voltage will then be determined by the input voltage minus the voltage drop across the internal P-MOSFET and the inductor.

## Over Current Protection (OCP) & Short-circuit protection function

The ME3120 provides over current protection and short-circuit protection function. When the load is too large and reaches the current limit point, the output voltage will decrease. When the output is shorted to ground or  $V_{FB}$  lower than 0.2V, the system works enter hiccup mode, to prevent current runaway from occurring. The soft start function will be triggered again every approximately 4.5mS. After the short circuit state is relieved, the system will automatically restart and return to normal working state.

## Output fast discharge resistor

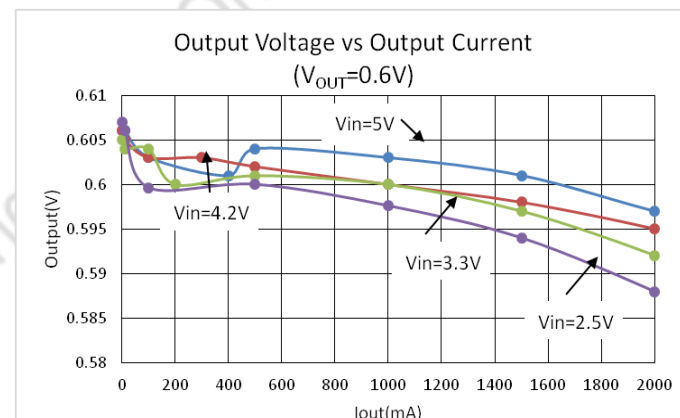
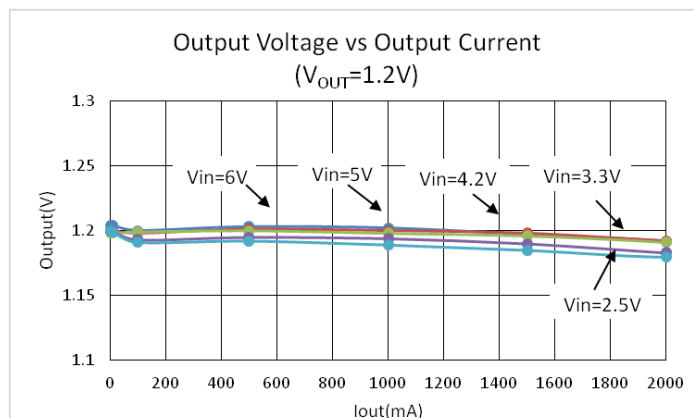
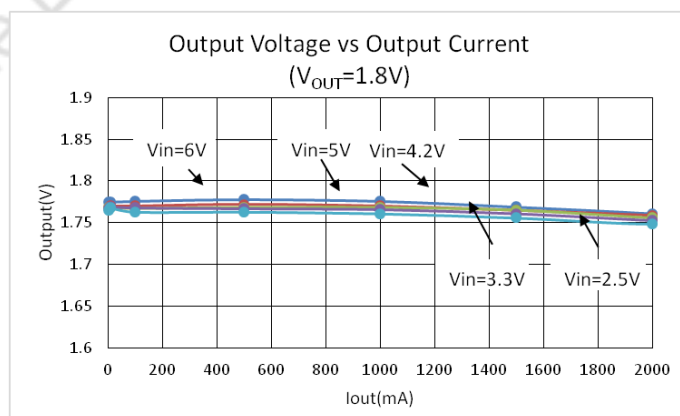
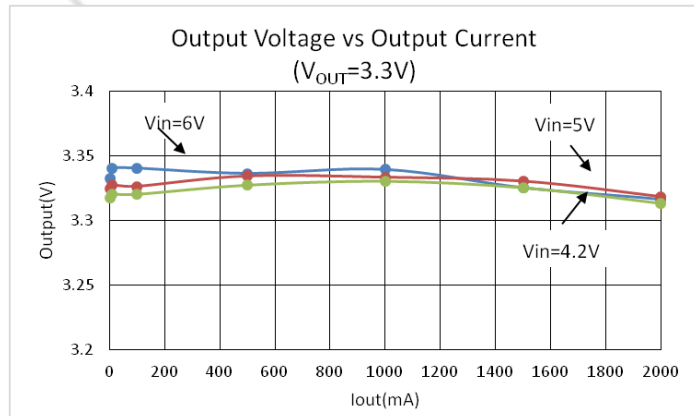
When  $V_{EN}=0V$ , the fast discharge path inside the chip is opened, and it discharges to ground through the SW pin, with a discharge impedance of about 220  $\Omega$ .

## Output Voltage Setting

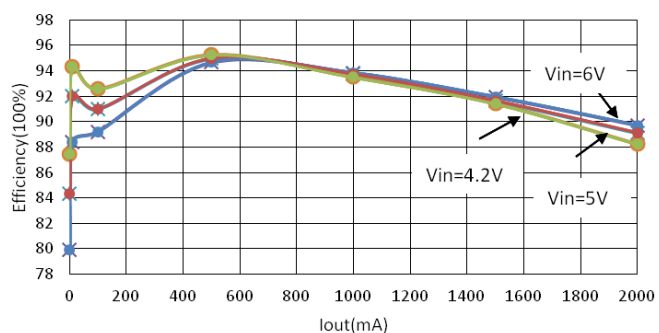
The output voltage is set by an external resistive divider according to the following equation:

$$V_{OUT}=V_{REF} \times (1+R_1/R_2)$$

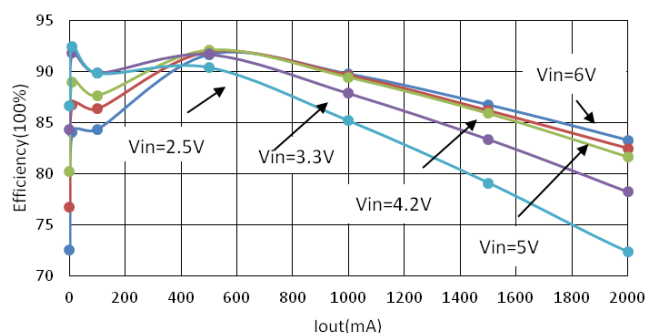
## Typical Performance Characteristic



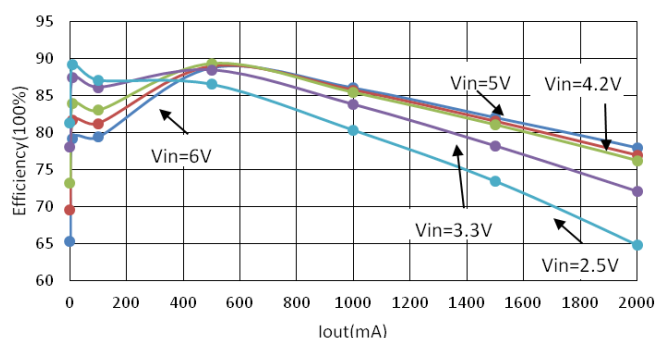
Efficiency vs Output Current( $V_{OUT}=3.3V$ )



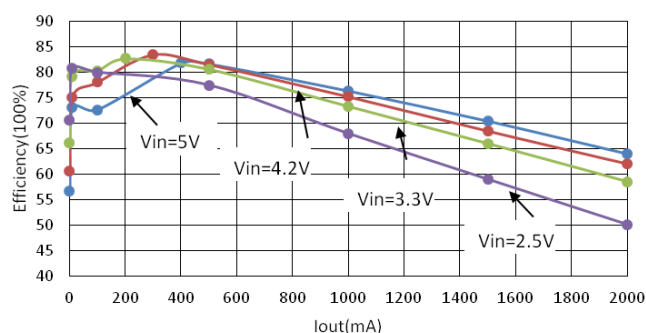
Efficiency vs Output Current( $V_{OUT}=1.8V$ )



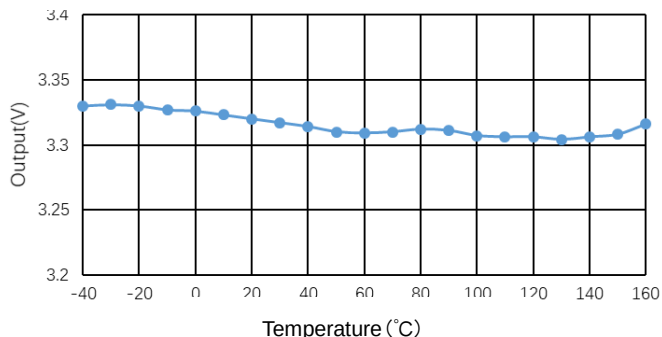
Efficiency vs Output Current( $V_{OUT}=1.2V$ )



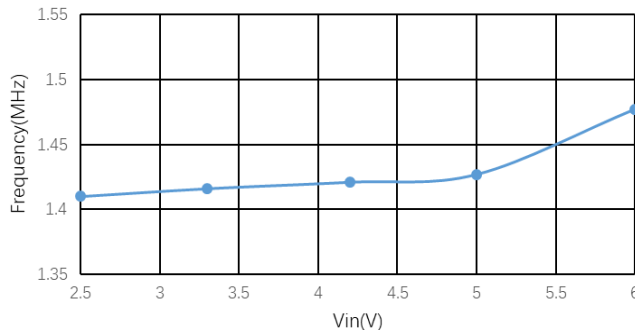
Efficiency vs Output Current( $V_{OUT}=0.6V$ )



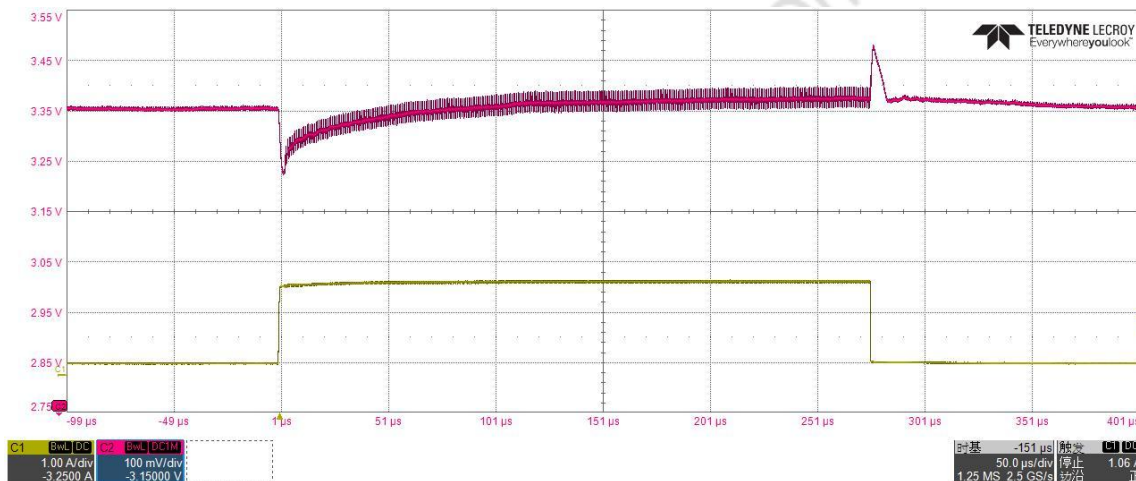
Output Voltage vs. Temperature



Frequency vs. Supply voltage

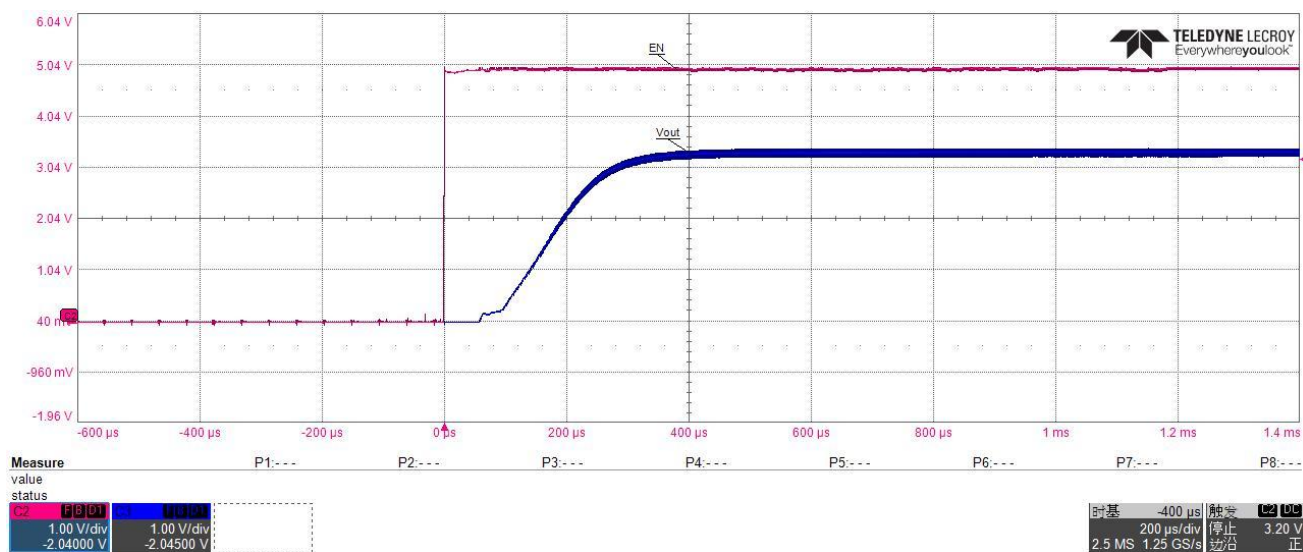


瞬态响应:  $I_{OUT}=230mA \rightarrow 1.85A \rightarrow 230mA$ ,  $V_{IN}=5V$ ,  $V_{OUT}=3.3V$ ,  $L=2.2\mu H$ ,  $C_{IN}=10\mu F$ ,  $C_{OUT}=22\mu F$ ,  $R1=1.1M\Omega$ ,  $R2=240k\Omega$ ,  $C_{FB}=47pF$

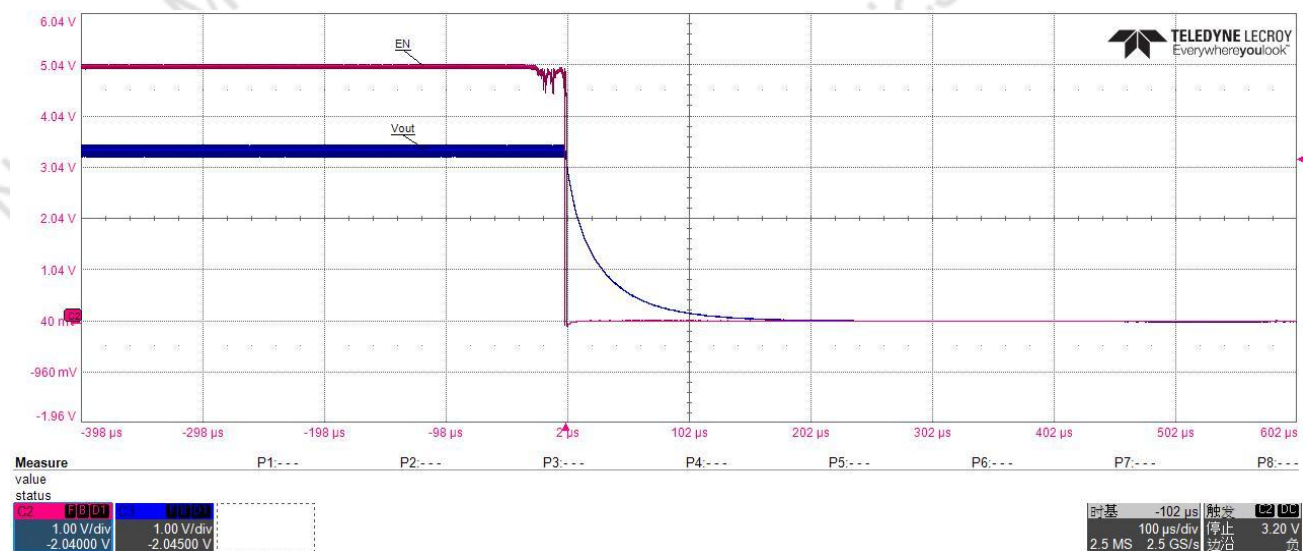




EN上电启动,  $V_{OUT}=3.3V$   $V_{IN}=5V$   $I_{OUT}=2A$



EN去电关闭,  $V_{OUT}=3.3V$   $V_{IN}=5V$   $I_{OUT}=2A$



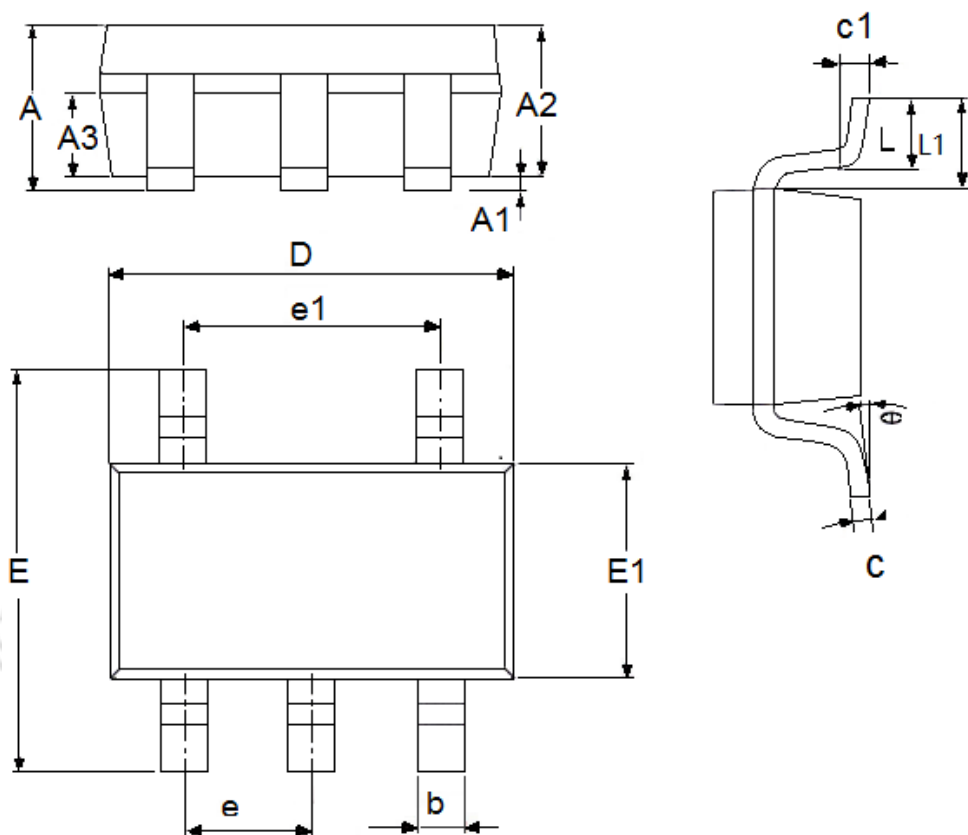
## Package Quantity

| Package Type | Minimum Packing QTY | UNITS       | Small Box | Large BOX |
|--------------|---------------------|-------------|-----------|-----------|
| SOT23-5      | 3000                | Tape & Reel | 30K       | 120K      |



## Package Information

- Package type: SOT23-5



| DIM | Millimeters |      | Inches      |        |
|-----|-------------|------|-------------|--------|
|     | Min         | Max  | Min         | Max    |
| A   | 1.05        | 1.45 | 0.0413      | 0.0571 |
| A1  | 0           | 0.15 | 0.0000      | 0.0059 |
| A2  | 0.9         | 1.3  | 0.0354      | 0.0512 |
| A3  | 0.6         | 0.7  | 0.0236      | 0.0276 |
| b   | 0.25        | 0.5  | 0.0098      | 0.0197 |
| c   | 0.1         | 0.23 | 0.0039      | 0.0091 |
| D   | 2.82        | 3.05 | 0.1110      | 0.1201 |
| e1  | 1.9(TYP)    |      | 0.0748(TYP) |        |
| E   | 2.6         | 3.05 | 0.1024      | 0.1201 |
| E1  | 1.5         | 1.75 | 0.0512      | 0.0689 |
| e   | 0.95(TYP)   |      | 0.0374(TYP) |        |
| L   | 0.3         | 0.6  | 0.0118      | 0.0236 |
| L1  | 0.59(TYP)   |      | 0.0232(TYP) |        |
| θ   | 0           | 8°   | 0.0000      | 8°     |
| c1  | 0.2(TYP)    |      | 0.0079(TYP) |        |

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